

EDUCATE Portal

Decentralized Education & Certificate Verification Platform

Project Documentation & Architecture Analysis

Introduction

What is the EDUCATE Portal?

- A comprehensive platform bridging traditional learning and modern blockchain technology.
- Designed to offer seamless course management for administrators.
- Empowers students by securely minting their academic achievements (certificates, ID cards, and semester results) onto the Web3 ecosystem.

Problem Statement

- **Vulnerability:** Traditional educational certificates and identity cards are highly susceptible to forgery, tampering, and physical damage.
- **Inefficiency:** Verifying the authenticity of paper-based or even conventional digital documents involves tedious, manual processes that are time-consuming for employers and institutions.
- **The Need:** A critical requirement for a decentralized, secure, and instantly verifiable system for academic credentials to ensure absolute trust and transparency.

Core Features

- **Admin & Candidate Dashboards:** Tailored portals for comprehensive management and student access.
- **Dynamic Result Management:** Customized workflows for Certificate (2 subjects) vs Diploma/Degree (Semester-based) courses.
- **Blockchain-based Verification:** Cryptographic hashes of documents are securely stored on Ethereum.
- **Decentralized Trust:** Third-party verification via QR codes and immutable ledger queries, preventing unauthorized modifications.

Project Flow

1. **Course & Batch Setup:** Admins create courses (Certificate or Diploma) and initialize batches.
2. **Candidate Onboarding:** Students sign up, and admins approve and assign them to active batches.
3. **Exam Scheduling & ID Requests:** Admins schedule exams; candidates request blockchain ID cards.
4. **Evaluation & Result Minting:** Admins upload marks. Semester results are anchored to the blockchain.
5. **Final Certificate Issuance:** Upon successful completion of all required semesters, a final consolidated certificate is minted.
6. **Real-time Verification:** Verifiers scan QR codes to confirm document authenticity via smart contracts.

Architecture Used

Hybrid Client-Server + Decentralized Web3 Architecture

- **Frontend (Client):** React.js, Vite, and Bootstrap provide a dynamic and responsive user interface, abstracting blockchain complexities.
- **Backend (Server):** Node.js and Express.js serve as the robust API layer for data handling and smart contract interaction via Ethers.js.
- **Database Layer:** MongoDB (with Mongoose ODM) manages structured, scalable data like student records and results.
- **Blockchain Layer:** Ethereum network (via Hardhat/Solidity) serves as the immutable ledger for storing transaction receipts and cryptographic hashes.

Project Diagram (Data Flow)

flowchart TD

Admin([Admin])

Student([Student/Candidate])

System((EDUCATE Portal \n Backend & DB))

BC[(Ethereum Blockchain)]

Admin -- Uploads Results & Schedules Exams --> System

Student -- Requests ID & Views Courses --> System

System -- Hashes Data & Interacts via Web3 --> BC

BC -- Transaction Receipt (Immutable) --> System

System -- Sends Confirmed Records --> Student

Database, Server, and Blockchain Tools

- **Database:**
 - *MongoDB* (Atlas/Local NoSQL database)
 - *Mongoose* (Object Data Modeling)
- **Server:**
 - *Node.js* (JavaScript Runtime)
 - *Express.js* (Fast, minimalist web framework)
- **Blockchain:**
 - *Solidity* (Smart contract programming language)
 - *Hardhat* (Local Ethereum development environment)
 - *MetaMask* (Crypto wallet gateway for admin transaction signing)
 - *Ethers.js* (Library for seamless backend-to-blockchain communication)

Project Repository & Screenshots

GitHub Repository:

<https://github.com/saikatssg/student-education-portal.git>

*(Insert screenshots of the platform below or view them directly on the GitHub repository
Readme)*

Landing Page (About Us)



Landing Page Placeholder

Admin Panel: Schedule Exams



Schedule Exams Placeholder

Admin Panel: Result Upload (MetaMask Integration)



MetaMask Integration Placeholder

Admin Panel: Result Anchored Successfully



Result Success Placeholder

Thank You!

The EDUCATE Portal

Securing the Future of Academic Credentials on the Blockchain.